Magnitude of a Vector

Exam code: 0607

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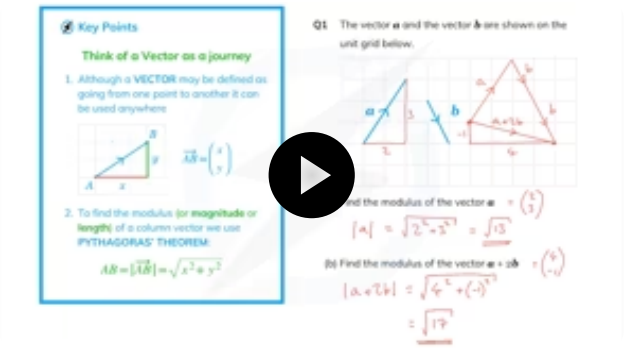

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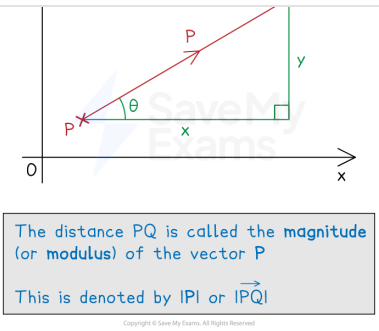
Magnitude of a vector

How do I find the magnitude of a vector?

- The **magnitude** of a **vector** is its **length** (distance)
  - It is also called the **modulus**
  - This is always a **positive** value
  - The direction of the vector is irrelevant
- The magnitude of $\overrightarrow{AB}$ is written $|\overrightarrow{AB}|$
  - The magnitude of **a** is written |**a**|
- Depending on the use of the vector, the magnitude of a vector represents **different quantities**
  - For velocity, magnitude would be speed
  - For a force, magnitude would be the strength of the force (in Newtons)
- In **component** form, the magnitude is the **hypotenuse** of a right-angled triangle
  - Use **Pythagoras' theorem** to find the magnitude
  - The magnitude of $\mathbf{a} = \begin{pmatrix} x \\ y \end{pmatrix}$
    - $|\mathbf{a}| = \sqrt{x^2 + y^2}$

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Examiner Tips and Tricks

- If there is no diagram, sketch one!
  - You can sketch a vector and use it to form a right-angled triangle

Worked Example

Consider two points $A(-3, 5)$ and $B(7, 1)$.

(a) Write down the column vector $\overrightarrow{AB}$.

Answer:

Find the horizontal and vertical distances between the two points
Subtract the *x* and *y* components of *A* from *B*

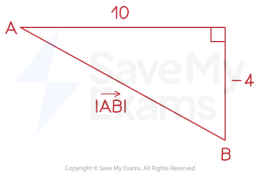
$$\overrightarrow{AB} = \begin{pmatrix} 7 - -3 \\ 1 - 5 \end{pmatrix}$$

$$\overrightarrow{AB} = \begin{pmatrix} 10 \\ -4 \end{pmatrix}$$

(b) Find the modulus of vector $\overrightarrow{AB}$.

Answer:

Sketching a diagram of the vector $\overrightarrow{AB}$ can help



Apply Pythagoras' theorem to the *x* and *y* components of $\overrightarrow{AB}$

$$= \sqrt{116}$$

$$|\overrightarrow{AB}| = 2\sqrt{29}$$

(c) Briefly explain why $|\overrightarrow{BA}| = |\overrightarrow{AB}|$.

Answer:

The magnitude of a vector is it's 'size'

Direction of the vector is ignored

$|\overrightarrow{BA}| = |\overrightarrow{AB}|$ **since both vectors have the same distance**

Another vector, $\overrightarrow{CD}$, has three times the magnitude of vector $\overrightarrow{AB}$.

(d) Write down a possible column vector for $\overrightarrow{CD}$.

Answer:

Being three times $|\overrightarrow{AB}|$ means the vector $\overrightarrow{AB}$ is three times longer

One way to find a vector is to multiply each component of the vector $\overrightarrow{AB}$ by 3 or -3

$$\overrightarrow{CD} = 3\overrightarrow{AB} = 3\begin{pmatrix} 10 \\ -4 \end{pmatrix}$$

$$\overrightarrow{CD} = \begin{pmatrix} 30 \\ -12 \end{pmatrix} \text{ or } \overrightarrow{CD} = \begin{pmatrix} -30 \\ 12 \end{pmatrix}$$

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